

HYPOTHESIS TESTING (PART 1)

18th August 2026

WHAT'S THE SIGNIFICANCE?

We have previously looked into parameter estimation, and in particular, point and interval estimation, to estimate unknown population parameters optimally. However, we may wish to assess a specific claim about an unknown parameter. The basic philosophy behind hypothesis testing is that rather than estimating a parameter, we claim that a proposed statement about the population is true and use sample data to assess whether the evidence supports or contradicts that statement. By quantifying the likelihood of observing the data under a particular assumption, we make objective decisions, in the presence of uncertainty, whether there is enough evidence to reject or not to reject the initial statement.

BASIC TERMINOLOGY

A statistical hypothesis test is a method of making statistical decisions using experimental data. In a hypothesis testing procedure the main task is to choose between two competing hypotheses about the distribution from which the experimental data are generated. Usually people put more faith on the hypothesis that they think is more likely, or sometimes, easier to describe. It is called the **null hypothesis**, denoted by H_0 . The competing hypothesis is called the **alternative hypothesis**, and is denoted by H_1 , sometimes H_A . The decision of a hypothesis test is either to accept or to reject the null hypothesis based on the value of a statistic, $T(X_1, X_2, \dots, X_n)$, computed on the basis of the sample, which is called the **test statistic**. Whenever the null hypothesis is rejected, the alternative hypothesis is accepted. Do note that the plausibility of the alternative hypothesis is never checked. It is accepted whenever it is decided that the null hypothesis is unacceptable. Thus, the alternative hypothesis should be chosen with care.

When a hypothesis considers one completely specified distribution, it is a **simple hypothesis**; otherwise, it is a **composite hypothesis**. The null hypothesis is rejected when the value of $T(X_1, X_2, \dots, X_n) \in R$, where R is such a set that $T(X_1, X_2, \dots, X_n)$ is not very likely to take values when H_0 is correct. There are two types of errors which can occur when a hypothesis is tested. This is a type I or type II error.

	H_0 is actually true	H_0 is actually false
Decision is taken to reject H_0	Type I error	
Decision is taken to accept H_0		Type II error

The probability of the type I error for a test is called the level of the test or the **significance level** of the test, denoted by α . The probability of the type II error is denoted by β . Thus, the power of a test is defined as $1 - \beta$, the probability of rejecting the null when it is false. The p-value is the probability of obtaining a result at least as extreme as the one that was actually observed when the null hypothesis, H_0 , is true.

Ideally a test should have $\alpha = \beta = 0$, i.e., there should not be any error. But that is, in general, not possible. Usually one of them increases when the other decreases. It is common practice to set an upper bound for α and then try to maximise power. The test with maximum power for a fixed level, α , is called the most powerful, MP, test at level α . When the alternative hypothesis is composite, if a test has maximum power against every alternative, it is called the uniformly most powerful, UMP, test at level α . Do note that UMP tests do not always exist.

LIKELIHOOD RATIO TESTS

Likelihood ratio tests are performed on the basis of the ratio of the likelihood under the distributions of the data under the null hypothesis and the alternative hypothesis. Thus, the likelihood ratio is defined as $\Lambda(X_1, X_2, \dots, X_n) = \frac{l(H_0|X_1, X_2, \dots, X_n)}{l(H_1|X_1, X_2, \dots, X_n)}$. The basic idea is that the likelihood would be higher for the true hypothesis, and thus, the likelihood ratio would be large if the null hypothesis, H_0 , is false, and small if H_1 is false. We reject H_0 if $\Lambda < c$ for some arbitrary constant, $c > 0$. Likelihood ratio tests are optimal for simple null versus simple or one-sided alternatives.

What happens if we look at composite nulls? Assume that the null hypothesis is now composite, and therefore, looks like $H_0 : \theta \in \Omega_0$, where Ω_0 is a multi-element subset of Ω , the space of all possible values of the parameter, θ . Consider $H_1 : \theta \in \Omega_0^c = \Omega_1$, where Ω_0^c is the complement of the set Ω_0 . For composite hypotheses, the likelihoods are evaluated for θ that maximise the likelihood under the corresponding hypothesis. That is $\Lambda^*(X_1, X_2, \dots, X_n) = \frac{\max_{\theta \in \Omega_0} l(\theta|X_1, X_2, \dots, X_n)}{\max_{\theta \in \Omega_1} l(\theta|X_1, X_2, \dots, X_n)}$. The test rejects H_0 for the small values of Λ^* . For calculations, we usually deal with $\Lambda(X_1, X_2, \dots, X_n) = \frac{\max_{\theta \in \Omega_0} l(\theta|X_1, X_2, \dots, X_n)}{\max_{\theta \in \Omega} l(\theta|X_1, X_2, \dots, X_n)}$, where $\Lambda = \min(\Lambda^*, 1)$. Thus, small values of Λ correspond to small values of Λ^* .

DUALITY OF CONFIDENCE INTERVALS AND HYPOTHESIS TESTING

Hypothesis tests and confidence intervals are interlinked with each other. The simple idea is that if the acceptance region of a test with null hypothesis $H_0 : \theta = \theta_0$ at level α is the set $A(\theta_0) = \{(X_1, X_2, \dots, X_n) : \theta_0 \in C(X_1, X_2, \dots, X_n)\}$, then the set $C(X_1, X_2, \dots, X_n) = \{\theta : (X_1, X_2, \dots, X_n) \in A(\theta_0)\}$ is a $100(1 - \alpha)\%$ confidence region for θ , and vice versa.

TESTS FOR GOODNESS OF FIT

Consider n IID observations $X_1, X_2, \dots, X_n$ from an unknown distribution F . Suppose that we want to test whether F is a certain suspected distribution, F_0 . Set the null as $H_0 : F = F_0$, and the alternative as $H_1 : F \neq F_0$. One way to perform this test is to group the observations into a number of classes, each of which is expected to contain at least a few observations. Let the number of observations in the i^{th} group be O_i . Let the expected number of observations in that group under F_0 be E_i . The statistic $\chi_0^2 = \sum_{\text{all cells}} \frac{(O_i - E_i)^2}{E_i}$, called Pearson's χ^2 statistic, is approximately chi-squared, with degrees of freedom ($d.f.$) = number of cells (k) - number of parameters fitted under $H_0(p)$ - 1 when H_0 is true. We can use χ_0^2 as a test statistic to test H_0 against H_A rejecting the null if χ_0^2 is large. Thus, the critical region of the test is $\chi_0^2 > \chi_{d.f.; \alpha}^2$. Do note that this test is not UMP.

TESTING OF HYPOTHESES

In hypothesis testing, we typically follow three simple steps, that is:

1. Set up the hypothesis test - Define the null and alternative hypotheses.
2. Calculate the test statistic - Based on different types of tests, we may use different methods.
3. Conclude - At the $\alpha\%$ significance level, there is (sufficient) evidence that we reject (or do not reject) H_0 .

CONCLUSION

Hypothesis testing provides a means to evaluate claims using observed data. By quantifying the likelihood of observing the data under a particular assumption, we make objective decisions whether there is enough evidence to reject or not to reject the initial statement.

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